

Performance Testing

Date	29 September 2026
Team ID	SWTID-2026-8574
Project Name	LegalEase: AI-Powered Legal Document Generator
Maximum Marks	5 Marks

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Testing Tool Used	Apache JMeter 5.6, Locust
Type of Testing	Load Testing, Stress Testing, Spike Testing
Target Module / API	Document Generation API (/api/generate), Auth Module (/api/auth/login)
Test Environment	Hosted Server - Render Cloud
Test Date	29 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Normal Load - Rental Agreement generation	50	60	Response <2s, PDF OK
2	Peak Load - 100 users on AI generation API	100	120	System stable <3s
3	Stress Test - 200 users doc generation	200	180	Error <5%, find breakpoint
4	Spike Test - 10 to 150 users sudden spike	150	60	System recovers without crash

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass/Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.45 seconds	Pass	Within SLA
2	Response Time (Max)	< 5 seconds	4.10 seconds	Pass	Max during stress
3	Throughput (Req/sec)	> 40 req/sec	52.3 req/sec	Pass	Good performance
4	Error Rate	< 1%	0.65%	Pass	Very low errors
5	CPU Utilization	< 80%	74%	Pass	Normal usage
6	Memory Utilization	< 80%	68%	Pass	Normal usage

Step 4: Observations & Analysis

Key Findings: LegalEase handled 100 concurrent users efficiently. Avg response 1.45s for document generation. Throughput 52.3 req/sec achieved. System stable up to 150 users. Bottlenecks Identified: 1. OpenAI API latency 4.1s during 200-user stress 2. PDF generation CPU spike 3. MongoDB template history query slow without indexing. Optimization Steps Taken: 1. Redis caching for templates 2. BullMQ async queue for PDF 3. MongoDB indexing for userId, templateId 4. Rate limiting for OpenAI API After optimization, avg response improved from 2.1s to 1.45s.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here - JMeter Summary Report]
